NITCI301: CCTV SYSTEM INSTALLATION

Sector: ICT and Multimedia
Trade: NETWORKING AND INTERNET TECHNOLOGIES
Module Type: Specific
Curriculum: ICTNIT3001- TVET Certificate III in networking and internet technologies
SCHOOL YEAR:2024-25

Competence: Install CCTV system

Learning Unit 1: Plan CCTV system installation

Learning Unit 2: Deploy CCTV equipment

Learning Unit 3: Configure CCTV System

Learning Unit 4: Operate CCTV system

Learning Unit 5: Test CCTV System

Learning Unit 6: Maintain CCTV system

Learning outcomes :1. Plan CCTV system installation

- **Introduction to CCTV system**
  - ✓ Definition

CCTV (closed-circuit television) is a TV system in which signals are not publicly distributed but are monitored, primarily for surveillance and security purposes. However, a CCTV setup is actually required to prevent crime. The very presence of a CCTV camera and it being noticed by a stranger convey a message to the person.

- ✓ **Working principles**

CCTV Cameras work by capturing a constant sequence of images that are then transmitted by cable or wirelessly to a display monitor where the sequence of images is demonstrated as video footage.

- ✓ **CCTV System main purpose**

- Crime Prevention.
- Monitors activities
- Collect evidence
- Keep records
- Real-time surveillance.
- Enhanced employee productivity.
- Reduces employee related incidences.
- Handling disputes.

- Enhance customer experience.
- Reduce Security costs.
  - **CCTV camera system is classified into the following classes:**
    1. Indoor

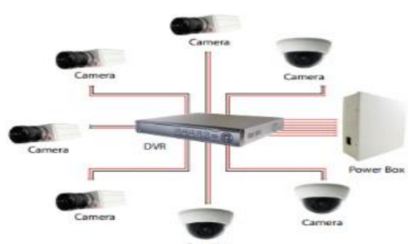
    The cameras can be smaller, more lightweight and are usually less intrusive than bulkier outdoor cameras. Both indoor and outdoor cameras utilize features like infrared, allowing for clear pictures in low light conditions and easy transitions when there is a sudden change in lightchanging automatically from color images in bright light to black and white when it gets darker.
  - **2 .Outdoor**

    Outdoor cameras are also more vulnerable to being tampered with, so they are typically made of more durable materials, like metal, and may be heavier or even housed in a casing in order to discourage easy removal.

✓ **Types of CCTV camera**

- Analog CCTV camera

  i.Analog CCTV Camera. standard definition cameras that use the composite video baseband signal (CVBS) to transmit video over a coaxial wire.
  Or An analogue camera is a traditional camera used in CCTV systems. It sends video over cable to VCRs or DVRs.



ii. (Internet Protocol) camera

An Internet Protocol camera, or IP camera, is a type of digital video camera that receives control data and sends image data via an IP network. An IP security system broadcasts their video as a digital stream over an IP network such as a WAN, LAN, Internet or Intranet. IP cameras combine the capabilities of cameras and some PC functionalities so that they don't require a direct connection to a PC to operate.

- Types of Digital/IP CCTV camera

An IP camera, or internet protocol camera, is a type of digital security camera that receives and sends video footage via an IP network.

Diagram					
Appropriate name	**A.Day / night**	**B.Bullet**	**C.Network**	**D.Dome**	**E.PTZ**

- ✓ **CCTV system components**

- Cameras.
- Lenses.
- Mountings and covers.
- Communication media such as cables.
- Power supply and power cables.
- Switching and synchronizers.
- Monitors.
- Video cassette recorders

- **Identification of CCTV system installation requirements**
  - ✓ **Introduction of Site survey**

- **Site Planning process**

Initial environment evaluation

Initial environmental examinations describe the environmental condition of a project, including potential impact, formulation of mitigation measures, and preparation of institutional requirements and environmental monitoring.

- ✓ **Selection of Analog and Digital phone device deployment**

Environment evaluation

- **Site surveys** are inspections of an area where work is proposed, to gather information for a design or an estimate to complete the initial tasks required for an outdoor activity. It can determine a precise location, access, best orientation for the site and the location of obstacles.

✓ **Types of Site survey**

- **Physical site survey**

A physical site survey is performed by walking the intended location and using site survey tools to measure the RF from an access point set up specifically for the survey. The access point will be moved to several locations intended to have coverage and the measurements taken there as well.

- **Passive physical site survey methodology**

Types of wireless site surveys

- **Predictive surveys** are performed before moving into a new space.
- **Passive surveys** collect information about all the signals in the environment after the site is built.
- **Active surveys** focus on a specific signal or set of signals while the network is in full operation.

During a passive survey, a site survey application passively listens to WLAN traffic to detect active access points, measure signal strength and noise level.
This used to be the most common method of pre-deployment Wi-Fi survey.

Location Survey: The location survey of a site is a determining factor for the type of CCTV installation for a site.
The location survey, in turn depends on the following factors:
- **Contents**
- **Risk**
- **Supervision levels**
- **Building**

- **Active survey methodology**

The study of algorithms is central to computer science and is of great practical importance to survey data analysis because algorithms are used in statistical programs.
Analysis of existing system

- **Current network usage**

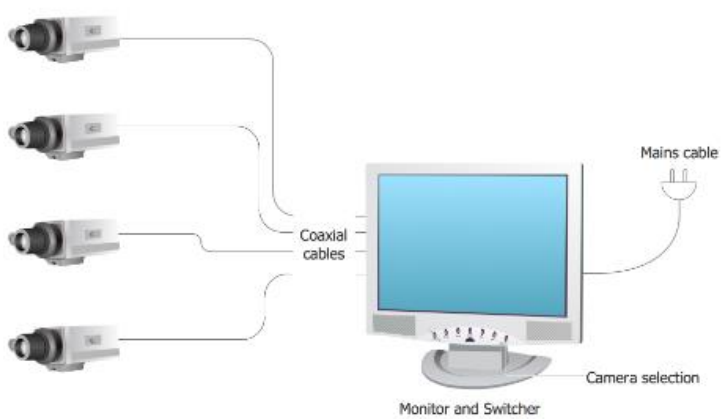
Network Usage displays the amount of data send and received by your network interfaces. Usage data is displayed in an easy-to-read bar graph.

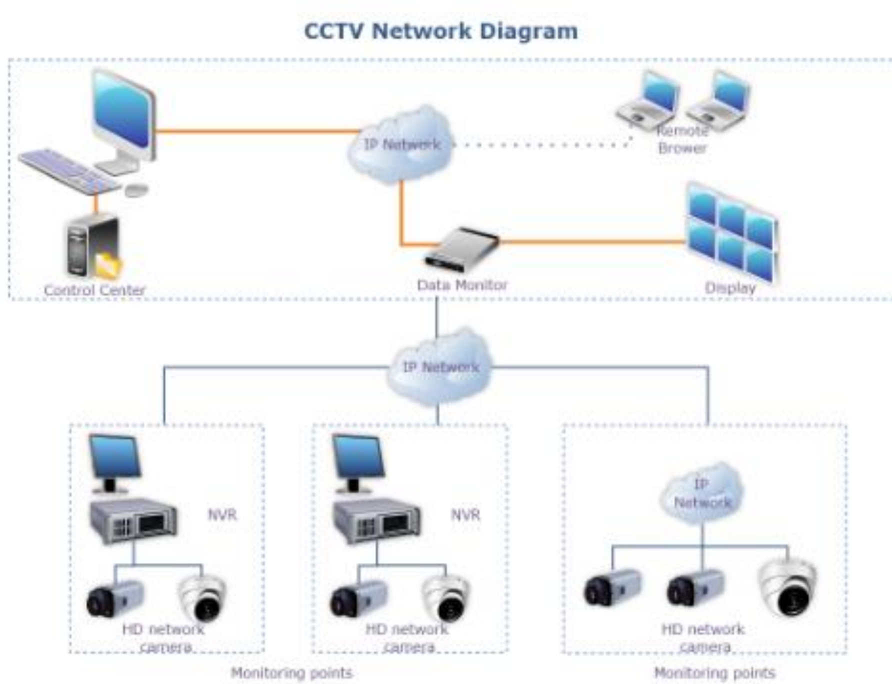
- **Future network usage**

The paper finds that the three *network* externality factors have different effects on the users' *future usage* intention for the four *Internet* services. Local *network*.

- **Design of CCTV Diagram**
    - ✓ **Description of CCTV Diagram**

A CCTV network diagram shows the closed-circuit television and how it is connected to the camera and the network to record the live feed.





- ✓ **The importance of designing a CCTV system Diagram**
- The CCTV diagram provides video cameras placement strategy
- increase security.
- CCTV diagram should include the scheme of strategic placement of video cameras, which capture and transmit videos to either a private network of monitors for real-time viewing, or to a video recorder for later reference.
- ✓ **Use of CCTV Diagram Design software**
- **Estimation of CCTV System Cost**

With professional installation, your setup cost could be higher than the cost of the camera. When installing, the contractor considers the following in their labour rates:

- ✓ Running cables through walls
- ✓ Eliminating signal interference
- ✓ Ensuring a clear view of the desired coverage area
- ✓ Installing cameras in a visible location to help act as a deterrent, but out of reach and not pointing at lights.
- ✓ Securing cameras and infrastructure
- ✓ Configuration of motion and privacy zones, notifications, video quality
- ✓

- ❖ **Items specification**
- ❖ **Items cost**
- ❖ **Total cost**

LEANING Outcome2: Deploy CCTV EQUIPMENT

- **Selection of tools, material and equipment of CCTV installation**

Tools: a device or implement, especially one held in the hand, used to carry out a particular function.

i.e: Screw drivers, Hammer…

✓ **Tools**

- **Cutting tools**

a cutting tool or cutter is typically a hardened metal tool that is used to cut, shape, and remove material from a workpiece by means of machining tools as well as abrasive tools by way of shear deformation.

- **Drilling tools**

A drilling machine is a power tool that is used to create cylindrical holes in a workpiece.

- **Fixing tools**

This metal tool is specifically designed to tighten cable ties and snip off the excess tail.

- Patching tools

- Testing tools

A cable tester **is a device that is used to** test **the strength and connectivity of a particular type of** cable **or other wired assemblies.**

- **Equi**pment

  - **Computer**

A **computer** These are used for viewing images and videos captured by the camera.

Today's computers have both kinds of programming.

  - **Monitor**

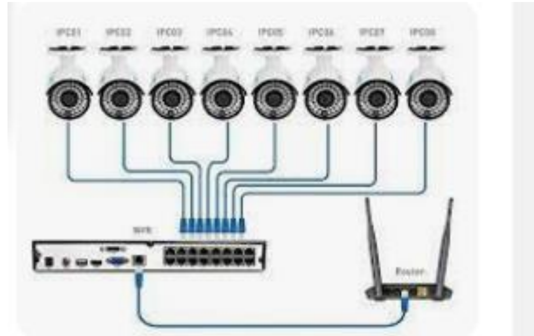
A computer monitor **is an output device that** displays information **in pictorial or textual form.**

  - Power Supply

A power supply is an electrical device that supplies electric power to an electrical load.

  - **Switch**

A network switch is a computer networking device that connects devices on a computer network by using packet switching to receive, process and forward data to the destination device.



- **Storage devices**

Nearly all digital cameras use removable storage like secure digital or SD memory cards for cameras

In addition to USB drives, flash memory devices also include SD and memory cards, which you'll recognize as the storage medium used in digital cameras.

- Cameras

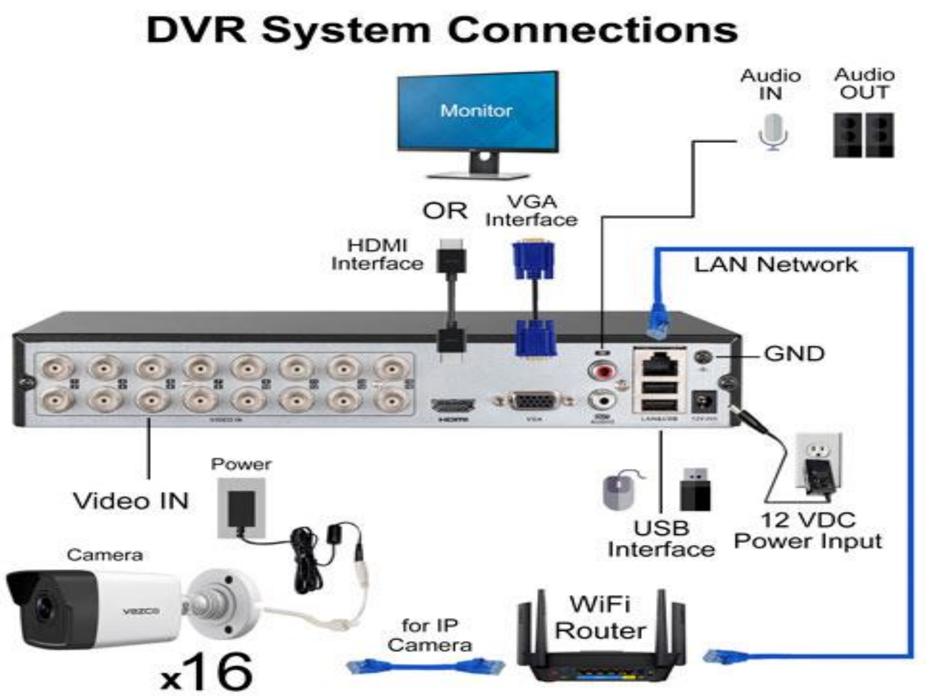
Camera, in photography, device for recording an image of an object on a light-sensitive surface.

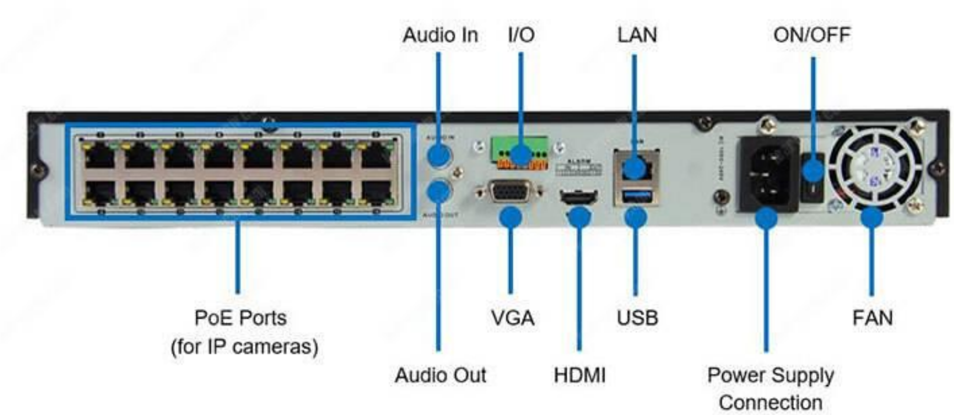
- **Recording Devices (NVR/DVR)**

A DVR converts analog footage into a digital format, while an NVR typically only works with digital footage. **DVR is** a Digital Video Recorder, and **NVR** is a Network Video Recorder.

Both NVRs and DVRs record video data transmitted from security cameras, storing it in a format that you can access later.

NVR: The NVR full form is network video recorder. As the name suggests, NVR recorders record videos from the network directly using Cat5 or Cat6 Ethernet cables with RJ45 plugs. NVR systems record and store video footage directly from the network it lives on. These systems work with an advanced type of camera, called IP cameras. IP cameras can actually capture and process video and audio data themselves. They do so using either an ethernet cable or wirelessly via an existing WiFi network.





- ✓ **Material**

- Cable ties

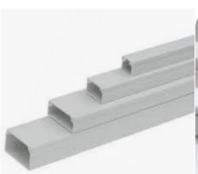
A cable tie is a band or length of strap, manufactured from a class of polymeric materials known as polyamides (Nylon 6/6)

Cable ties organize all kinds of cables, like those used with a computer, an entertainment system, or in a network.

- Cable trunks

Cable trunking is an enclosure usually with a rectangular cross section, and with one removable or hinged side, that is used to protect cables and provide space for other electrical equipment.

- Pipes

To convey substances which can flow — liquids and gases (fluids), slurries, powders and masses of small solids.

- Screws

Screws are to hold objects together and to position objects.

- Wall Mounts

A wall-mounted **television can save space and make a room look better.**

- **Power extension**

Is a length of flexible electrical power cable (flex) with a plug on one end and one or more sockets on the other end **(usually of the same type as the plug).**

- Labelling tags

Labels are die-cut plastics, papers, metals, or other materials that can be affixed to containers or surfaces.

- CCTV Power supply box

Allows surveillance system installers to easily manage the power to multiple CCTV cameras at a central point (usually at the location of the DVR).

- **Conducting CCTV Trunking**
    - ✓ Description of installation types
        - Visible
        - **Built in**

forming an integral part of a structure or object. a camera with a built-in flash. especially: constructed as or in a recess in a wall. a built-in bookshelf.: built into the ground**.**

- **Semi built in**

Semi-integrated dishwashers offer controls that are built into the exterior panel of the dishwasher door so that the controls are visible when the dishwasher is closed.

- **Water proof**

impervious to water. especially: covered or treated with a material (such as a solution of rubber) to prevent permeation by water.

- **Underground**

Keep your property secure

- **Identification of Trunking Materials types**
  - **Plastic**

This initiative uses several innovative digital tools to identify plastic hotspots along the streets and waterways.

  - **Stainless steel**

These stainless-steel cameras are protected against weather and corrosion. Find cameras, adapters, and housings that can withstand extreme environments.


- **Wooden**

Wooden Camera designs, manufactures, and retails innovative camera accessories for professionals.

✓ **Application of cable laying and pulling**

- **General method**

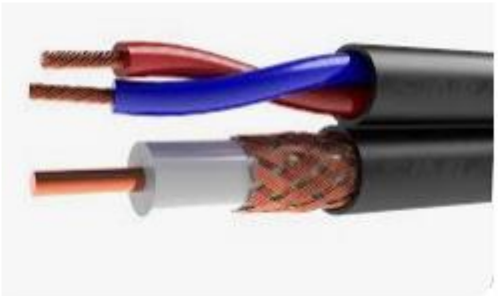
Direct buried cables shall be taken to ensure that the laying area is not subject to landslide or cracking, that there are no obstacles in the subsoil and there is no soil contamination.


Cable laying arrangement

Analog, HD-SDI and HD-CVI cameras will need two wires run to them. One for video transmission and a set of power wires in order to power the camera.

Segregation of cables

different parts of the same electrical installation. circuits operating at different voltages. circuits supplying different safety services. safety services and the remainder of the electrical installation.



- **Assembling CCTV equipment**
  - ✓ **Rack mount**

Camera mount is the mounting part on the lens side for fixing to the camera. Each camera mount has a different size thread and flange back (FB: distance from sensor to lens flange)Common camera mount FB are shown below

- ✓ **Camera**

- Mark the spot where you'll attach the bracket for mounting.
- Attach and secure the camera.
- Install the DVR (Digital Video Recorder) in a secure spot.
- Plan the cable pathways from the cameras to the DVR.
- Connect the network cable to the internet router.
- Connect each camera to the DVR or NVR.
- Once everything is hooked up and powered on, check the video feed to see if the cameras are capturing their intended areas.

✓ **Recording devices (NVR (**Network Video Recorder**)/DVR (**Digital Video Recorder

As the diagram shows, **DVR** systems rely on a coaxial cable to transmit the raw data, or unprocessed video signal. This signal is not digital, and only becomes so when it is encoded in the digital video recorder. Once encoded, it can be stored or viewed.

In an NVR system, cameras like fixed IP cameras or bullet IP cameras can record and process video data within the device itself. It is a digital video recording camera that can also handle audio. This encoded video and audio data is sent over WiFi or an ethernet cable to the NVR recorder, where it is stored for decoding and remote viewing.

- **Fixing and connecting CCTV controller devices and camera**
  - ✓ **Terminating cables**
  - ✓ **Patching**
    - **Patch panel types**

- **Twisted-pair copper.** These panels are designed for specific twisted-pair copper specifications, like Cat5E, Cat6, Cat6A and Cat7 cables.
- **Fiber optics.** Patch panels are available for both single- and multimode fiber cabling.
- **Coax**. Coaxial cable is most commonly used for AV installations.

Patch code types

Patch cables can be made from a variety of cable types:

- Coaxial
- UTP
- STP
- fiber

Patch panel management types

patch panel is one special kind of patch panel. They are useful tools to build a neat and easy-to-manage cabling system.

materials and their use

- **Soldering tin:** is a joining process used to join different types of metals, usually made of tin and lead which is melted using a hot iron and cooled to create a bond. Soldering tin is a metal or metallic alloy used when melted to join metallic surfaces especially an alloy of lead and tin so used. 
- **Glue** is a sticky substance used for joining things together, often for repairing broken things. 

- **Silicone glue is** a type of adhesive that contains silicon and oxygen atoms, making it a good water-resistant solution.

- **Insulator tape:** Electrical tape (or insulating tape) is a type of pressure-sensitive tape used to insulate electrical wires and other materials that conduct electricity. It can be made of many plastics, but vinyl is most popular, as it stretches well and gives an effective and long lasting insulation.

- **A screw is** a type of fastener, in some ways similar to a bolt typically made of metal, and characterized by a helical ridge, known as a male thread (external thread).

- **Universal anchors:** It is a plastic used to put in hole before you insert screw to make sure screw is very ties.

- **Trunking is** used to protect cables from damage and to hide unsightly cables from view. It can be used in almost any location including homes, hotels and hospitals. Trunking is available in many materials, sizes and forms to suit the location and requirements of the installation**.**

Learning Outcome 3: Configure CCTV System

- **Management of Administration parameters settings**

CCTV Camera installation parameters include its height, vertical and horizontal orientation, frequency, resolution and focus points of the images being acquired.

- ✓ **Change default password**

1) Login to your CCTV system via computer browser (Chrome, Firefox, or IE), or the CMS software you have installed.

2) If you log in using browser, go to "configuration" > "System" > "User Management", you will see user details over there.

- ✓ **set recording storage**

SD Card Storage: Many modern CCTV cameras come with in-built storage memory. The storage is quite limited, so an SD card becomes necessary. You can save the particular footage on your SD card. Along with video servers, they prove helpful as a backup system**.**

In general, properties keep CCTV footage as long as they have the storage capacity to do so. In most cases, this means supermarkets and other properties keep footage for around 30 to 90 days. However, this varies a great deal. Some only keep footage for a few hours while others may retain it indefinitely.

- **Management of CCTV user accounts**

- ✓ **Identification of user account**

Management **software should** use **an administrator** account **with its own credentials to access the cameras. This** account **should be unique and not shared.**

For DVR and NVR, following are the default accounts:

1. admin/admin admin/123456 (master account of local and network accessing)
2. 888888/888888 (local administrator)
3. 666666/666666 (restricted user)
4. default / default (hidden user)

- ✓ **Creation of user account**

Step by Step Instructions

1. Log into the web interface of the device.
2. Click the Menu icon in the top left of the page.
3. Select Account.
4. Click Add User.
5. The Add User menu will appear
6. In the Authority menu you can define the individual rights of the User using the System, Playback, and Monitor tabs as well as which channels the user can access.
7. A message will confirm when the user is successfully added

How do I set my CCTV IP address?

1. Use a Command Prompt with the Ipconfig command. The easiest way to find out the IP address scheme of a network is by using the Command Prompt program.
2. Change the network settings on the PC to a 192.168.
3. Login to the camera and configure it for the main network scheme.
4. Change the computer network settings back to default.

- ✓ **Setting of factor authentication**
  - **Knowledge factor authentication**

A knowledge factor is the most popular category of credentials used for authentication. Often referred to as “something you know,” this authentication requires the user to provide knowledge on confidential information before accessing a secured network

- **Two factor Authentication**

- **Configure CCTV IP addresses**

Network configuration

- Connect the network camera to your Local Area Network (LAN). A LAN is often the home network of a consumer. ...
- Find the IP address of the network camera. There are a few ways to do it. ...
- Fix the IP address (i.e. make the IP address static). ...
- Start using the Wi-Fi. ...
- Find the new IP again.

✓ Static IP

How To Set a Camera to Static IP

1. Step 1: Power on the device and on a laptop or PC, access the web UI for your device.
2. Step 2: Once you have logged into your web UI, navigate to the TCP/IP menu of your device by going to, Setup>>Network>>TCP/IP.
3. Note: The IP address located in the IP Address field is now set as static.

✓ **Dynamic IP**

CCTV DDNS is a service that enables you to access the security cameras remotely each time over the Internet with the dynamic IP address.

How to set static IP address for CCTV camera?

1. Use a Command Prompt with the Ipconfig command. The easiest way to find out the IP address scheme of a network is by using the Command Prompt program. ...
2. Change the network settings on the PC to a 192.168. ...
3. Login to the camera and configure it for the main network scheme. ...
4. Change the computer network settings back to default.

- **Apply setting of CCTV recordings**

- ✓ video format

What is the format of CCTV video?

Compression techniques are used in Digital CCTV to reduce the file sizes of recorded video images. Typical compression formats used for video are: **MJPEG, MPEG4** & H. 264.
.

- ✓ **overwrite capability**

When CCTV footage is recorded, it is saved on a local hard disc, or a cloud server, or an offsite server. In most cases, after 15 days or a month depending upon the storage available in the DVR/NVR, **old data is overwritten by fresh data by default, and thus old data is no longer available.**

Fortunately, the answer is yes! You can use several methods to recover any lost, overwritten, or deleted CCTV surveillance video.

What is overwritten video?

When a video is overwritten, what does this mean? Overwrite means to record or copy new data over existing data when a file or directory is updated.

- ✓ **backup frequency**

What is backup frequency?

Backup frequency is a management class attribute
Currently, the majority of wireless security cameras operate on the 2.4 GHz frequency

Do CCTV cameras have backup?

There is a USB port on the front of the DVR into which you insert the memory stick. Using the on-screen menu and supplied mouse you first search for footage and then make a backup directly onto the memory stick.

- ✓ **set time and date**

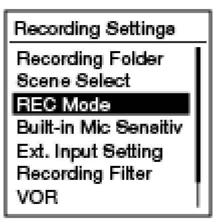
camera go into setup. system general and then under date and time it pulls up here in the general tab make sure you click on date and time.

- ✓ **set recording mode**

You can set the recording mode for the files to be recorded. Set this menu item before you start recording.

1. Select BACK/HOME - " Settings" - "Recording Settings" - "REC Mode," and then press .

   When you set the recording mode for FM radio recording, select BACK/HOME - " Settings" - "FM Radio Settings" - "REC Mode (FM Radio)," and then press (ICD-UX543F/UX544F only).



2. Select the desired recording mode, and then press .
3. Press and hold BACK/HOME to return to the HOME menu.

   Press STOP to return to the window displayed before you entered the HOME menu.

- **Document the CCTV functionalities**

keep track of the interior and exterior of a property, transmit the signal to a monitor or set of monitors, and give real-time 24/7 viewing access.

- ✓ **Introduction to documentation**

Documentation is any communicable material that is used to describe, explain or instruct regarding some attributes of an object, system or procedure, such as its parts, assembly, installation, maintenance and use.

A CCTV camera system makes use of video cameras, also called surveillance cameras to keep track of the interior and exterior of a property, transmit the signal to a monitor or set of monitors, and give real-time 24/7 viewing access.

- ✓ **Cctv system requirements**
  - **Tools, material and equipement**

Tools You Need for CCTV Installation

- Ladder
- Power drill
- Drill bits – including masonry bits
- Electric heavy-duty screwdriver
- Small screwdrivers
- Pencil and pen
- Hammer
- BNC crimp tool
- Cable stripper
- Cutter and Stanley knife
- Pliers
- Safety goggles, helmet, boots
- Cable Rod Set & Fish Taps to feed cables in awkward places
- Electrical tape to stick cables to rod
- Labels to mark cable
- Raw plugs
- Electrical meter to check electricity
- Video test monitor to plugin directly to camera, to adjust angle of camera
- Network test tool

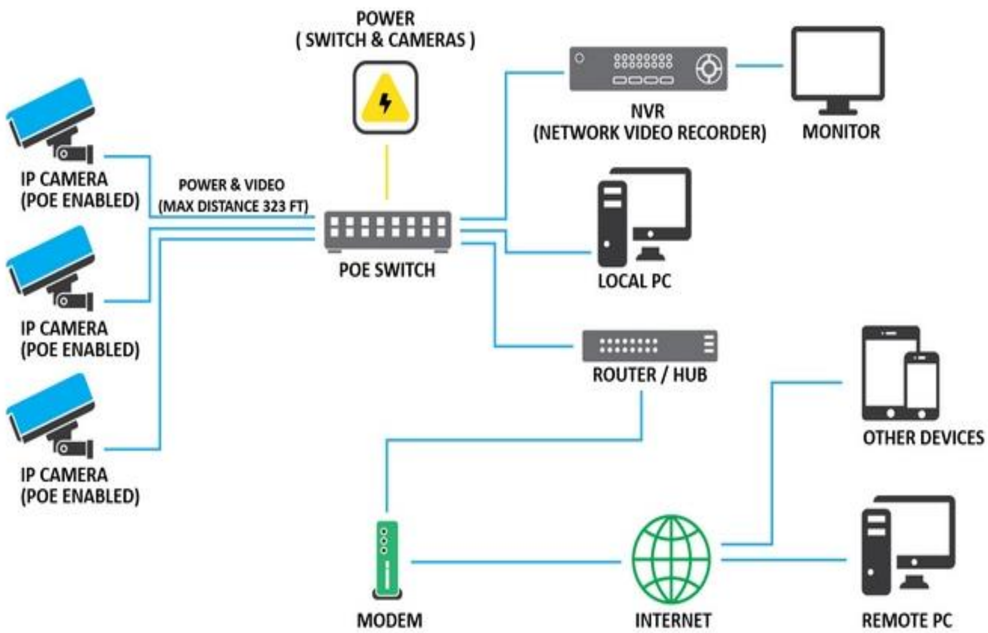
What equipment is needed for CCTV?

A CCTV Camera System has become a vital necessity in this modern world. While selecting your preferred security system, the 5 essential components you need to give importance to for consistent service are: Camera, Monitor, Cable, Video Recorders, and Data Storage.

5 Essential Components of CCTV Camera System

- Camera. If you're building a CCTV Camera System, you have two camera options: Internet Protocol (IP) or analog.
- Monitoring Station.
- Cables & Routers.
- Video Recorders.
- Data Storage.

- **Cctv diagram**



- **cost**

✓ **CCTV system settings**

How do I set up my CCTV system?

Here is a quick and simple CCTV Installation Guide.

1. Define your security needs and choose the equipment.
2. Choose suitable spots for cameras and DVR/NVR.
3. Mount your DVR/NVR and cameras.
4. Connect your cables and monitor.
5. Test your CCTV system.

LEARNING Outcome4: Operate CCTV system

- **Monitor live footage of capture events**
  - ✓ **Definition of live footage**

Live Footage means live audio-visual or visual-only footage of any and all Matches during the Event, including the period from the first ball to the last ball of any Match and shall extend from the pre-match toss until the end of the post-match presentation ceremony; Sample 1Sample 2.

  - ✓ **Type of monitors for CCTV**

Modern CCTV monitors are available in various types, such as:

- LED monitors.
- LCD monitors.
- TFT monitors.

  - ✓ **viewing live footage for CCTV**

You can view CCTV cameras live in the local area networks (LAN), which is typically within your home area, with all types of IP cameras.

- Have No WiFi Network? Try 4G LTE Cameras
- Want Local Live Stream Without Network? Try IP Cameras
- Want PC/TV Live Video Without Network? Get an NVR

- **Maintain CCTV system control room**
  - ✓ **Describe control room**
    - **Definition**

A control room is typically a large space where the operations of a business or industry can be monitored or controlled via remote systems.

    - **Function of the control room**

is a room that serves as a central control and monitoring station for security, safety, building and other types of control room system in a large physical facility or area.

- **Control room operator**

Control Room Operators work in the control rooms of large plants, in particular power plants, where they monitor all of the operations of the plants to ensure that everything is working properly.

Control Room Operators control the creation and flow of electricity from power plants to businesses, homes and factories.

- ✓ **Environmental condition**
  - **Control room arrangement**

A CCTV control room is usually a central hub within a facility or organization where security personnel control and monitor security systems and processes. Setting them up mainly aims to monitor activities occurring inside a building and protect people and businesses from potential threats

- **Air condition system**

An air conditioning system, or a standalone air conditioner, provides cooling and/or humidity control for all or part of a building.

- **Electricity**

What is CCTV camera power consumption in watts? The power consumption of a typical CCTV camera ranges from 4 to 15 watts, depending on the type. While the DVR device that is connected to the CCTV camera uses between 10 and 40 watts.

- **Equipment status**

Equipment status indicates the current state of a piece of equipment.

- Camera. If you're building a CCTV Camera System, you have two camera options: Internet Protocol (IP) or analog.
- Monitoring Station.
- Cables & Routers.
- Video Recorders.
- Data Storage.

- **Perform** Export/Playback of required information
  - ✓ **Define Export and Playback**

What is Export?

The process of sending a file out through a specialized mini-application or plug-in, so as to print or compress it**.**

Playback

An act or instance of reproducing recorded sound or pictures often immediately after recording. Play back.

  - ✓ Performing steps of Export and Playback for required information

- **Interpretation of CCTV system logs**
  - ✓ **Definition of logs**

CCTV Logbook is a secure portal that enables CCTV system owners to manage their CCTV system and assets more efficiently.

  - ✓ **Description of system logs metrics**

Unlike an event log, which records specific events, metrics are a measured value derived from system performance**.**

Generate and interpret incident report

To use the CCTV equipment in the shared services area of observation to help detect and deter crime and anti-social behaviour and provide high quality evidence to allow enforcement agencies to investigate and prosecute offenders.

How to write a security report

1. Take notes. Details and observations make up the bulk of your security reports. ...
2. Start with a summary.
3. Detail the narrative.
4. Follow the form.
5. Proofread.
6. Avoid emotional language.
7. Avoid abbreviations and conjunctions.
8. Be prompt

What are skills of CCTV operator?

Skills and qualities

- To be honest, responsible, reliable and alert.
- Good observational skills.
- Excellent communication skills, including clear speech.
- The ability to deal calmly with emergency situations.
- Common sense, curiosity and a practical approach.
- To be able to keep clear written records.

Learning Outcome 5: Test CCTV System

- **Check deployed CCTV system**

A single security camera tester to power, configure and document. SecuriTEST IP CCTV tester is an installation and troubleshooting tester for digital/IP, HD coax and analog security camera systems.

  - ✓ Material
  - ✓ Equipment

- **Selection of testing tools**

- ✓ **hardware testing tools**

How To Test Your Security Camera System In 3 Simple Steps

1. Step 1: Check Camera Positioning and Focus. The first step in testing your CCTV system is to check the positioning and focus of your cameras. ...
2. Step 2: Check Image Quality. ...
3. Step 3: Test Recording and Playback Functions.

- ✓ **software testing tools**

Common CCTV Problems and Fixes

1. Some Common CCTV Problems with Fixes.
2. Verify camera power and connections.
3. Discover and ping the camera.
4. Reboot camera. ...
5. Be Aware of username/password.
6. Verify no IP conflict.
7. Check ARP tables.
8. Upgrade firmware.

- **Performing connectivity testing**
  - ✓ Physical connectivity
  - ✓ Logical connectivity
- **Performing Functional testing**
  - ✓ Hardware
  - ✓ Software
- **Testing performance**
  - ✓ Visual quality

Generally speaking, security cameras have a range of anywhere between 10 and 70ft during the day. Alternatively, some night-vision security cameras have a range of 100-200ft! There are many features that can affect and impact your security camera's vision range, such as: Focal length.

- ✓ **Latency**

Latency is the time between the instant a frame is captured and the instant that frame is displayed.

- ✓ **Audio quality**

Regardless of whether the mic is built-in or external, security camera microphones are sensitive enough to pick up sounds from a 40ft radius when in a quiet room.

- ✓ **NVR throughput**

An average 1080p camera will use about 5Mbps for 30FPS video*, but many NVRs only allow about 2Mbps per camera.

- ✓ **DVR throughput**

Depending on the camera's configuration and use, they may consume as little as 1 GB of data per month. The typical monthly data use of IP camera is about 60 GB.

- **Elaboration of testing report**
  - ✓ **Generate testing report**

Steps to create a good test summary report

1. Step 1: Capture the purpose of the document.
2. Step 2: Capture an overview of the product in test.
3. Step 3: Capture the Testing Scope.
4. Step 4: Capture the Metrics.
5. Step 5: Capture the types of testing performed.
6. Step 6: Capture the Test Environment and the Tools used.

- ✓ **Interpret testing report**

The objectives of the process of testing and commissioning of CCTV camera is to verify proper functioning of the equipment's and system after installation**,**

Learning Outcome 6: Maintain CCTV System

- Performing preventive maintenance
  - ✓ **Set periodical backup**
    - **Definition**

Periodic backups are performed on a set schedule, rather than event-driven, such as whenever new data is saved to a hard drive or existing data files are modified

- **Types.**

Full backup: The most basic and comprehensive backup method, where all data is sent to another location.

Incremental backup: Backs up all files that have changed since the last backup occurred.

Differential backup: Backs up only copies of all files that have changed since the last full backup.

- **Benefit**

- ➢ improve disaster recovery and, by extension
- ➢ business continuity
- ➢ improves data security and enables faster disaster recovery

- **Schedule backups**

Schedule backup is an IT activity that is cleverly used to automatically execute the backup script

- **Checking of camera positioning**

CCTV cameras should be positioned at all entrances and exits to the premises. This will give your client full visibility of who is coming and going at all times.

- **Check security measures**
  - **Physical security**

a system in which images are monitored and recorded for surveillance and security purposes.

  - **Logical security**

Anti-virus software, passwords, and encryption are all examples of logical security mechanisms.

- **Performing corrective maintenance**
  - Hardware maintenance
    - Identifications of common problems
    - Repair/replace damaged devices and cables
  - Software maintenance
    - Identification of common problems
    - Regular Change of controller device credentials
    - Installing updates and features of firmware and software for devices
- **Elaboration of maintenance report**
  - Introduction to Maintenance report

maintenance report is a document that holds specific data about inspections and tasks as well as their effects on overall maintenance operations.

- Elements of Maintenance report
- Status before maintenance
- Activity Procedures
- Tool, material used
- Status after maintenance
- Recommendations

…………………….END!!.......................................